

Building with Blueprints

The Quiz Master | Session 1

Hello, Future Coding Architect! Coding is like magic—you build your own world with simple blueprints! Today, we are moving beyond simple lines of code to become master architects of our own Quiz Game.

The Challenge of Manual Quizzing

Imagine trying to manage 100 questions one by one. Without a plan, your code becomes messy and hard to track. It's easy to lose track of which questions you've already asked!

The Manual Way:

Question: *"What is the national fruit of India?"*

Answer: *"Mango"*

Imagine writing this block 100 times! There must be a smarter way!

Class vs. Object: The Blueprint

In Python, we use a **Class** to solve our messy code problem. Think of it using the **Rubber Stamp Analogy**:

Stamp Design (CLASS)

The **Blueprint**. This is the idea or the design template. You create it once!

Ink Marks (OBJECT)

The **Real Thing**. Every time you press that stamp on paper, you get a real ink mark object!

Our 'Question' Blueprint

Let's write our first blueprint in Python code. This will be the design for every quiz question we make:

```
class Question:
    def __init__(self, text, answer):
        self.text = text
        self.answer = answer
```

- **class Question:** Tells Python we are making a new blueprint.
- **The `__init__` Method:** A special setup method that runs when creating each object.
- **Attributes (`self.text`, `self.answer`):** Information held by each object.

Creating & Running Our Quiz

Now that we have our stamp (the Class), let's make some ink marks (the Objects) and store them in a list:

```
q1 = Question("Capital of India?", "New Delhi")
q2 = Question("IPL 2023 winner?", "CSK")
q3 = Question("ISRO chief?", "Dr. S. Somanath")

# Storing our objects in a list
questions = [q1, q2, q3]
```

To run our game, we loop through our list and do three things for every question:

1. **Show Question Text:** Display the question on the screen.
2. **Get Answer:** Wait for the player to type their response using `input()`.
3. **Check & Respond:** Compare the player's answer to the real answer and say "Correct!" or "Try again!".

```
for question in questions:
    user_answer = input(question.text + " ")
    if user_answer.lower() == question.answer.lower():
        print("Correct!")
    else:
        print("Incorrect. Try next question.")
```

The Quiz Master Challenge!

Practice Worksheet | Grade 4 Coders

Test your knowledge of Python classes, objects, and how our quiz game works! Read each question carefully and write your answers. Good luck, Master Architect!

Question 1

In Python, a class is like a blueprint. What is the actual "real thing" we make from that blueprint called?

A) A Variable

B) An Object

C) A List

D) A Loop

Question 2

If a Class is a "Rubber Stamp Design", what are the "Ink Marks" printed on the paper?

A) Functions

B) Classes

C) Objects

D) Notebooks

Question 3

Which special method in Python is used to initialize (set up) the attributes of a new object?

A) `__start__()`

B) `__init__()`

C) `__make__()`

D) `__setup__()`

Question 4

Look at this code: `q1 = Question("Capital of India?", "New Delhi")`. What are the two attributes stored inside this object?

Question 5

What Python built-in function do we use to let players type in their answers on the screen?

A) print()

B) input()

C) type()

D) ask()

Question 6

Why do we use `.lower()` when checking our player's answer in the quiz code?

Question 7

If we have 100 questions, what tool in Python helps us go through them one by one without writing manual code for each?

A) An 'if' statement

B) A 'for' loop

C) A Class

D) A variable

Question 8

How do we store multiple question objects (like `q1`, `q2`, `q3`) together so we can manage them easily?

A) Inside a List []

B) Inside a String " "

C) Inside a loop

D) Inside a print statement

Question 9

True or False: We can only create one single object from a class blueprint.

A) True

B) False

Question 10

What keyword does Python use to start defining a new blueprint?

Question 11

If `user_answer = "mango"` and `question.answer = "Mango"`, will `user_answer.lower() == question.answer.lower()` be True or False?

A) True

B) False

Question 12

In our code, `self.text` is an attribute. What is the other attribute we created in our `Question` blueprint?
